

## SUMMARY TABLE: CHEMISTRY GRADE 10

### Sub-Strand 1.4: Chemical Bonding — Teacher Reference (Pre-filled)

| SUMMARY TABLE: CHEMISTRY GRADE 10 |                                                                                                                                                                                                                                                         |
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| Sub-Strand                        | Sub-Strand 1.4: Chemical Bonding                                                                                                                                                                                                                        |
| Driving Question                  | DRIVING QUESTION: Why do diamond and graphite — both made of only carbon atoms — have such completely different properties, and how does the way atoms bond together determine what a substance looks like, how it behaves, and what we can use it for? |

### INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

| Lesson # and Title                                                                                                   | What did I observe?                                                                                                                                                                                                                                                                                                                                                                                     | What did I learn?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | How does this explain the phenomenon?                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| <b>Lesson 1</b><br><b>The Carbon Paradox: Launching the Phenomenon</b>                                               | Students observed that graphite (pencil stick, labelled 100% Carbon) smeared a grey mark on a ceramic tile and lit a conductivity bulb brightly, while diamond (also labelled 100% Carbon) scratched and gouged the tile surface without being damaged, and did not conduct electricity at all. Students also viewed a silent animation showing the contrasting atomic arrangements of both allotropes. | Students learned that diamond and graphite are both made entirely of carbon atoms — the same element, same atomic structure — yet have completely opposite physical properties: hardness vs. softness, electrical insulator vs. conductor, transparent vs. opaque. Students learned that differences in properties between substances made of the same atoms must be caused by differences in how those atoms are arranged and bonded — not by differences in what atoms are present. Students also learned that writing initial models and posing genuine questions on a Driving Question Board are core scientific practices. | No bonding theory was explained in this lesson. The lesson was deliberately designed to open a knowledge gap — to make students feel the puzzle — rather than to resolve it. The full explanation (tetrahedral covalent network bonding in diamond vs. layered hexagonal covalent bonding with delocalised electrons in graphite) will be built across the unit as students learn about valence electrons, bond types, Lewis structures, and the relationship between structure and physical properties. |
| <b>Lesson 2</b><br><b>Valence Electrons &amp; the Octet Rule: The Electron Bookkeeping Behind the Carbon Paradox</b> | Learners completed an electron configuration table for 16 elements across Periods 1–3, colour-identified valence electrons in red, and noticed the periodic reset pattern from Period 2 to Period 3.                                                                                                                                                                                                    | Valence electrons are the outermost-shell electrons that participate in bonding; atoms gain, lose, or share electrons to achieve an octet (or duplet for Period 1) matching the nearest noble gas; carbon has exactly 4 valence electrons shown as 4 unpaired Lewis dots, priming it to form 4 bonds.                                                                                                                                                                                                                                                                                                                           | Carbon's 4 unpaired valence electrons mean it can form 4 bonds — the octet rule is the engine driving this. In diamond and graphite, carbon uses all 4 valence electrons in different bonding geometries, which is why two substances made of identical atoms behave so differently.                                                                                                                                                                                                                     |
| <b>Lesson 3</b>                                                                                                      | Students observed (1) electron transfer                                                                                                                                                                                                                                                                                                                                                                 | Students learned that: (1) ionic bonds form                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Students explained the Carbon Paradox                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

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| <b>Ionic Bonding: Electron Transfer and the Giant Ionic Structure</b>                                                     | <p>modelled physically using Na and Cl atom cards with moveable electron tokens, (2) the resulting formation of <math>\text{Na}^+</math> and <math>\text{Cl}^-</math> ions with complete outer electron shells, (3) a simulation of the giant ionic lattice showing alternating positive and negative ions in a 3-D arrangement, and (4) a physical model of the NaCl lattice cross-section built from dried maize (<math>\text{Na}^+</math>) and beans (<math>\text{Cl}^-</math>).</p>                                                                                                                                                                                                   | <p>when a metal atom transfers one or more valence electrons to a non-metal atom, satisfying the octet rule for both; (2) the transferred electrons produce oppositely charged ions (cations and anions) held by strong electrostatic attraction; (3) the giant ionic lattice of NaCl — with strong forces in all directions — explains its high melting point, brittleness, and conductivity only when dissolved or molten; (4) dot-and-cross diagrams are a standard tool to represent electron transfer in ionic bond formation.</p>                                                                                                                                                                                                                                                                         | <p>connection by noting that NaCl's giant ionic structure (electron transfer, charged ions, electrostatic lattice) is fundamentally different from both diamond's giant covalent network (electron sharing, no ions, directional covalent bonds) and graphite's layered structure (delocalised electrons, weak inter-layer forces) — reinforcing that the TYPE of bonding, not merely the presence of a giant structure, determines properties and uses.</p>                                                                                                                                                                            |
| <b>Lesson 4</b><br><b>Covalent &amp; Dative Covalent Bonding: Drawing Lewis Structures and Exploring Electron Sharing</b> | <p>Learners observed ARES simulation animations showing electron-cloud overlap forming single (<math>\text{H}_2</math>), double (<math>\text{O}_2</math>) and triple (<math>\text{N}_2</math>) covalent bonds, and completed an observation table recording shared pairs, lone pairs and noble gas configuration for each molecule. Learners also physically constructed dot-and-cross diagrams for <math>\text{H}_2\text{O}</math>, <math>\text{CO}_2</math> and <math>\text{CH}_4</math> using the Lewis Structure Construction Kit cards, and examined a Bonding Comparison Card showing partial Lewis structures of graphite's hexagonal layer and diamond's tetrahedral network.</p> | <p>Learners learned that covalent bonds form when atoms SHARE pairs of electrons (rather than transfer them as in ionic bonding), that carbon's 4 valence electrons allow it to form 4 covalent bonds, that dative (coordinate) covalent bonds occur when one atom donates BOTH electrons of a shared pair (as in <math>\text{NH}_4^+</math> and <math>\text{H}_3\text{O}^+</math>), and that in graphite each carbon forms only 3 covalent bonds leaving one electron per atom delocalised and free to move — explaining graphite's electrical conductivity — while in diamond all 4 bonds are used in a 3-D tetrahedral network leaving no free electrons, explaining why diamond is an insulator.</p>                                                                                                        | <p>Learners explained (in their group 3-sentence summaries and cumulative model panels) why graphite conducts electricity while diamond does not, using the concepts of bonding pairs, lone pairs, delocalised electrons and the octet rule — directly addressing the Carbon Paradox anchoring phenomenon and advancing the driving question: the way carbon atoms bond to one another, not the identity of the atoms themselves, determines all the observable properties of the substance.</p>                                                                                                                                        |
| <b>Lesson 5</b><br><b>Hydrogen Bonding &amp; Van der Waals Forces: Why Water Defies Expectation</b>                       | <p>Iodine crystals turned directly into purple vapour when gently warmed (sublimation with minimal liquid phase). The boiling point data showed water boiling at <math>100\text{ }^\circ\text{C}</math> — far above the expected trend extrapolated from <math>\text{H}_2\text{S}</math> (<math>-60\text{ }^\circ\text{C}</math>) and <math>\text{H}_2\text{Se}</math> (<math>-41\text{ }^\circ\text{C}</math>). Noble gases showed increasing boiling points with increasing atomic mass.</p>                                                                                                                                                                                            | <p>Water's anomalously high boiling point is caused by hydrogen bonding — a strong intermolecular attraction between the <math>\delta^+</math> hydrogen of one water molecule and a lone pair on the <math>\delta^-</math> oxygen of a neighbouring molecule. Hydrogen bonding requires H to be bonded to N, O, or F. Van der Waals forces include London dispersion forces (temporary dipoles in all molecules/atoms — stronger in larger, more polarisable electron clouds) and dipole–dipole forces (between permanently polar molecules). Iodine's relatively high boiling point despite being non-polar is explained by strong London dispersion forces from its large, polarisable electron cloud. Graphite's layers are held together by weak Van der Waals forces — this explains why layers slide,</p> | <p>The Carbon Paradox is now deeper: diamond has no intermolecular forces because it is a continuous covalent network — every carbon is bonded to four others in 3-D, so there are no 'separate molecules' and no layers. Graphite, however, HAS layers, and between those layers only weak Van der Waals forces act. This is precisely why graphite is soft (layers slide past each other easily) while diamond is the hardest natural substance (no weak links anywhere in the structure). Intermolecular forces are the 'hidden layer' of chemistry that controls boiling points, melting points, and physical states of matter.</p> |

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|                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | making graphite soft and able to write on paper.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Lesson 6</b><br><b>Metallic Bonding: The Sea of Electrons — From Copper Wire to Aluminium Foil</b>                                | Aluminium foil, copper wire and an iron nail could all be bent without breaking; all three metals lit the conductivity bulb when tested; solid NaCl and candle wax did not conduct electricity; metals feel cool to the touch and have melting points hundreds of degrees higher than simple molecular substances like iodine or water.                                                                                                                                                                                                         | Metallic bonding is the electrostatic attraction between a regular lattice of positive metal ions and a surrounding sea of delocalised electrons that are free to move throughout the entire structure and belong to no single ion.                                                                                                                                                                                                                                                                                                                                                      | Electrical conductivity in metals is explained by the free movement of delocalised electrons carrying charge; ductility and malleability are explained by the non-directional nature of the electron sea — ion layers slide without bond-breaking; relatively high melting points reflect the strong electrostatic attraction across the entire lattice; graphite conducts because one electron per carbon atom is delocalised within each layer, giving it partial metallic character despite being a covalent solid.                                                                                                                                                                                                                                                                           |
| <b>Lesson 7</b><br><b>Physical Properties Investigation — Part 1: Setup &amp; Prediction</b>                                         | Students examined four unknown solid samples (labelled A–D) using sensory observation and matched them against a set of published property Data Cards showing melting point ranges, solubility, and conductivity data for four structural types (labelled W–X–Y–Z), without being told the substance names. They also watched an ARES-platform particle-level animation showing NaCl (Mombasa salt flats context), sugar, sand, and aluminium dissolving or not dissolving in water.                                                            | Students learned that each structural type — giant ionic, simple molecular, giant covalent, and giant metallic — has a characteristic pattern of physical properties (solubility, conductivity, melting point) that arises directly from its bond type and particle arrangement. They learned that graphite is an anomaly within giant covalent structures because its delocalised electrons allow electrical conductivity, unlike diamond or SiO <sub>2</sub> . They practised using Claim–Evidence–Reasoning to distinguish particle-level explanation from surface-level observation. | Students explained their predictions by linking bond type to particle behaviour: ionic compounds dissolve because water molecules pull apart oppositely charged ions; simple molecular compounds have low melting points because only weak intermolecular (Van der Waals) forces must be overcome; giant covalent structures have very high melting points because strong covalent bonds throughout the lattice require enormous energy to break; metallic structures conduct electricity because delocalised electrons move freely through the lattice. Graphite's conductivity was explained as a consequence of its one-delocalised-electron-per-carbon-atom layer structure — the same structural feature that makes it fundamentally different from diamond despite both being 100% carbon. |
| <b>Lesson 8</b><br><b>Physical Properties Investigation — Part 2: Data Collection (Solubility, Conductivity &amp; Melting Point)</b> | Learners tested five substances (NaCl, sugar, graphite, SiO <sub>2</sub> sand, aluminium foil) at three investigation stations: (A) solubility in water, (B) electrical conductivity as solid and when dissolved, and (C) melting-point ranking from data cards. They recorded that NaCl dissolved fully and conducted in solution but not as a solid; sugar dissolved but did not conduct; graphite did not dissolve but conducted as a solid; SiO <sub>2</sub> sand did not dissolve and did not conduct; aluminium foil did not dissolve but | Learners identified that bond type appears to determine physical properties: ionic compounds (NaCl) conduct only when dissolved because ions become free to move; simple molecular compounds (sugar) do not conduct even in solution because they do not form ions; giant metallic structures (aluminium) conduct as solids because of free electrons; giant covalent structures (SiO <sub>2</sub> , diamond) do not conduct. They recognised that graphite is an anomaly within giant covalent structures —                                                                             | Learners updated their cumulative Bond-Type Property Model with evidence from all three stations, adding a column for physical property data and revising earlier entries where new evidence contradicted prior predictions. They posted yellow (evidence confirmed) and blue (new questions) notes on the Driving Question Board, clustering open questions around the carbon paradox. The model's Giant Covalent row now contains a two-cell diamond/graphite sub-table distinguishing their conductivity                                                                                                                                                                                                                                                                                      |

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|                                                                                                                  | <p>conducted as a solid. Melting-point data cards showed giant covalent and metallic structures at the top of the ranking. Diamond and graphite data cards were added to the melting-point chart — both near the top — yet graphite conducted and diamond did not.</p>                                                                               | <p>it conducts — and that this anomaly traces directly back to the anchoring phenomenon. Pattern Statements were recorded and partially completed, with the graphite/diamond distinction flagged as an open question for Lesson 9.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <p>results, with the explanation cell left intentionally incomplete and marked as the key unresolved question to be addressed in Lesson 9.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <p><b>Lesson 9</b><br/><b>Physical Properties Investigation Part 3: Analysis, Explanation and DQB Update</b></p> | <p>Students observed from their Lesson 8 data that NaCl conducts only when dissolved, sugar does not conduct at all, aluminium conducts in the solid state, SiO<sub>2</sub> does not conduct, and graphite — a giant covalent structure like SiO<sub>2</sub> — conducts electricity, making it an anomaly among the four structure types tested.</p> | <p>Students learned that the four structure types differ systematically in conductivity, melting point and solubility because of differences in the particles present and the forces or bonds between them: ionic lattices require free-moving ions for conduction; metallic lattices have permanently delocalised electrons; simple molecular substances have no ions or free electrons in any state; giant covalent structures generally have no free particles for conduction. Graphite is the critical exception: its three-bond-per-carbon arrangement leaves one delocalised electron per atom free to move along each layer, giving it metallic-like conductivity despite being a covalent structure of non-metal atoms.</p> | <p>The properties of a substance are determined not merely by which atoms are present but by how those atoms are bonded and arranged. Diamond and graphite both consist entirely of carbon atoms, yet their different bonding arrangements — four bonds in a 3-D tetrahedral network versus three bonds in a layered hexagonal structure with delocalised electrons — produce completely opposite properties: hardness versus softness, electrical insulation versus electrical conductivity, transparency versus opacity. This is the structural explanation for the anchoring phenomenon and the central answer to the driving question.</p> |
| <p><b>Lesson 10</b><br/><b>Resultant Structures &amp; Uses: From Bond Type to Real-World Application</b></p>     | <p>Students examined Substance Dossier cards containing physical-property data tables and Kenyan-use photographs for diamond, graphite, NaCl, aluminium, and SiO<sub>2</sub>, and conducted a gallery walk around six Kenyan-context material posters including a mystery substance (silicon carbide) used in a Kisumu grinding wheel.</p>           | <p>Students learned that the real-world use of a material is determined by its bond type and resultant structure: diamond's 3-D tetrahedral giant covalent network makes it the hardest natural substance (used in cutting tools); graphite's layered structure with delocalised electrons makes it a conductor and lubricant (used in dry-cell batteries and pencils); NaCl's ionic lattice makes it soluble and non-toxic (used in food preservation); aluminium's metallic lattice makes it malleable and conductive (used in Kenyan cookware and construction); SiO<sub>2</sub>'s giant covalent network gives it a very high melting point and transparency (used in glass-making).</p>                                        | <p>Students explained, through Claim–Evidence–Reasoning paragraphs and cumulative model annotations, how the bond type of each substance produces the specific physical properties that make it suitable for its Kenyan use — and extended this reasoning to a novel giant covalent substance (SiC), bringing the anchoring paradox (diamond vs. graphite) to a full structural resolution.</p>                                                                                                                                                                                                                                                |
| <p><b>Lesson 11</b><br/><b>Animations, Simulations &amp; Digital Literacy: Seeing Chemical</b></p>               | <p>Learners watched curated animations and simulations of ionic (NaCl lattice), giant covalent (diamond and graphite), and metallic (aluminium) bonding structures,</p>                                                                                                                                                                              | <p>Learners can explain how digital simulations visualise the 3-D tetrahedral network of diamond (no free electrons, no conductivity) and the layered hexagonal</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>The Carbon Paradox: diamond and graphite are both 100% carbon, but diamond's 3-D tetrahedral covalent network (no free electrons) makes it hard and non-</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

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| <b>Bonding in Motion</b>                                                                                            | and completed Simulation Critique Cards comparing digital models to their own Lewis diagrams and physical-property data from earlier lessons.                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | structure of graphite (delocalised pi electrons, conducts electricity), and can articulate at least one specific simplification or limitation of each digital model.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | conducting, while graphite's layered structure with one delocalised electron per carbon atom makes it soft between layers and electrically conducting — a difference that no simulation fully captures but that digital models make far easier to visualise and understand than a static Lewis diagram alone.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Lesson 12</b><br><b>Project Lesson — 3-D Model Building: Bonding Structures with Locally Available Materials</b> | Students physically constructed 3-D models of NaCl ionic lattice, graphite layered structure, diamond tetrahedral network, and SiO <sub>2</sub> giant covalent lattice using clay, wire, bottle tops, maize, string, and cardboard. During the carousel gallery walk, students observed each other's completed models and recorded peer critiques on Peer Review Cards. The anchoring phenomenon was recreated when the diamond and graphite models were placed side by side on the display shelf — echoing the original Lesson 1 display of the real diamond chip and pencil graphite on the teacher's desk. | Students learned that the three-dimensional geometry of chemical bonds — 109.5° tetrahedral in diamond and SiO <sub>2</sub> , 120° trigonal planar within graphite layers, and the alternating octahedral arrangement in the NaCl lattice — is the direct structural cause of every physical property the class has investigated across Lessons 1–11. Specifically: all four covalent bonds in diamond are used for bonding (no free electrons → insulator; rigid 3-D network → hardest natural substance); graphite's third bond delocalises one electron per carbon (→ conductor) while layers are held only by Van der Waals forces (→ soft, lubricating); NaCl's ionic lattice gives high melting point but allows conduction when molten; SiO <sub>2</sub> 's continuous covalent network gives extreme hardness and a very high melting point. The lesson also embedded the PCIs of environmental conservation (recycled local materials) and creative thinking (model design). | The Carbon Paradox driving question can now be answered structurally: diamond and graphite are both pure carbon, but in diamond every carbon forms four covalent bonds in a continuous 3-D tetrahedral network — every electron is bonded, the structure is rigid in all three dimensions, and there are no free electrons. In graphite, every carbon forms three covalent bonds in flat hexagonal layers, leaving one electron per atom delocalised across the layer; the layers themselves are held together only by weak Van der Waals forces. This single difference in bonding geometry explains hardness (diamond: bonds resist deformation in all directions; graphite: layers slide), electrical conductivity (diamond: no free electrons; graphite: delocalised electrons carry charge), opacity and lustre, and every other property contrast observed in Lessons 1–11. The 3-D models built today make this invisible nano-scale geometry tangible and directly observable. |
| <b>Lesson 13</b><br><b>Model Presentations, Driving Question Board Final Revision and Unit Consolidation</b>        | In this final lesson, we observed four 3-D models of chemical bonding structures — the NaCl ionic lattice, the graphite layered hexagonal structure, the diamond tetrahedral network and the SiO <sub>2</sub> continuous covalent network — built by student groups from locally available materials. We also observed the original anchoring phenomenon: the diamond chip and the graphite pencil stick, now viewed with thirteen lessons of chemical bonding knowledge. We observed all student-generated questions on the Driving Question Board being answered with                                       | We learned that the type of chemical bond formed between atoms — determined entirely by the number and arrangement of valence electrons — produces a specific three-dimensional structure that directly causes every physical property a substance has. Giant ionic structures (NaCl) have high melting points and conduct electricity only when molten or dissolved. Simple molecular structures (iodine, water) have low melting points because only weak Van der Waals or hydrogen bonds hold molecules together. Giant covalent structures (diamond, SiO <sub>2</sub> ) are extremely hard with very high melting                                                                                                                                                                                                                                                                                                                                                                 | We explained the Carbon Paradox fully: in diamond, each carbon atom forms four strong covalent bonds in a tetrahedral arrangement (bond angle 109.5 degrees) creating a rigid three-dimensional network with no free electrons — making diamond the hardest naturally occurring substance and a perfect electrical insulator. In graphite, each carbon atom forms only three covalent bonds in a flat hexagonal arrangement (bond angle 120 degrees) and the fourth electron from each carbon is delocalised along each layer — making graphite a good electrical conductor. The                                                                                                                                                                                                                                                                                                                                                                                                       |

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|  | evidence gathered across the unit. | points because every electron is used in strong covalent bonds throughout a continuous network. Giant metallic structures (aluminium, copper) conduct electricity and heat because delocalised electrons move freely. We also learned that two substances made of the same atoms — diamond and graphite — can have completely opposite properties because the bonding arrangement of those atoms is entirely different. | layers themselves are held together only by weak Van der Waals forces, so they slide over one another easily, making graphite soft enough to write with and useful as a lubricant. The bond type determines the structure, the structure determines the properties, and the properties determine the uses — from Nairobi jewellers cutting tools to pencils used in every Kenyan classroom. |
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## ATTRIBUTION AND LICENSING

**Curriculum Alignment:** Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

**Content & Pedagogy:** Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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